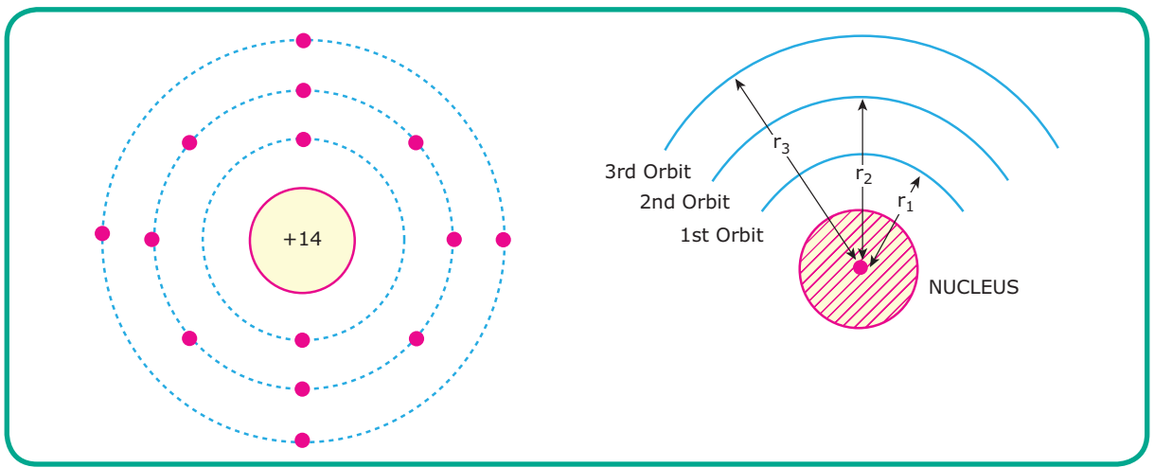


Figure3.12 Structure of silicon atom

Figure 3.12 shows the structure of silicon atom. It has 14 electrons, 2 in the first, 8 in the second and remaining 4 electrons in third orbit. The first, second, third orbits are also known as K, L and M orbits, respectively.

These electron can revolve only in permitted orbits (i.e., orbits of radii r_1, r_2 and r_3) and not in any orbit. Thus, all radii between r_1 and r_2 or between r_2 and r_3 are forbidden. Each orbit has fixed amount of energy associated with it. If an electron in the first orbit is to be lifted to the second orbit, just the right amount of energy should be supplied to it. When this electron jumps from second orbit to first, it gives back the acquired energy in the form of electromagnetic radiations.

3.5.2 Energy Level

It has already been discussed that each orbit has fixed amount of energy associated with it. The electrons moving in a particular orbit possess the energy of that orbit. The larger the orbit, greater is its energy. It becomes clear that outer orbit electrons possess more energy than the inner orbit electrons.

Figure 3.13 shows the energy of different orbits. This is one way of the representing the energy in orbits and is known as energy band diagram. The first orbit represents first energy level; the second orbit indicates the second energy level and so on. The larger the orbit of the electron, the greater is its energy and higher is the energy level.

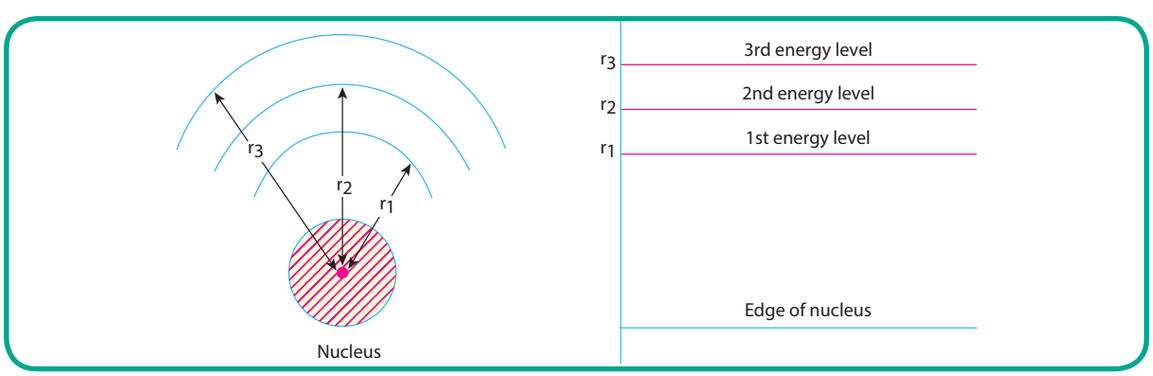


Figure 3.13 Energy level diagram

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